DocuMind Enterprise

Production-Ready Enterprise RAG System

Built with FastAPI, Groq, Pinecone, React

💡 **Created by:** Sarvesh Ghotekar
Role: AI / Full Stack Engineer
Focus: Production-ready RAG systems, AI safety, scalable architectures
Status: Active development & optimization

Project Overview

What is DocuMind Enterprise?

DocuMind Enterprise is a production-ready Retrieval-Augmented Generation (RAG) system designed for enterprises to extract intelligence from internal documents. It enables secure PDF ingestion, real-time AI chat with citations, and strict hallucination prevention.

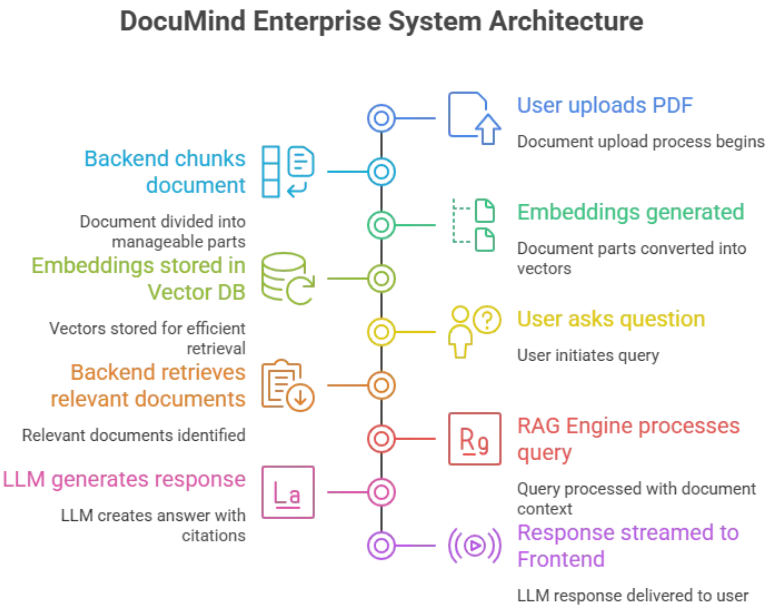
Who is this for?

- Mid to large enterprises
- Legal, policy, compliance, and knowledge teams
- Organizations requiring **auditable AI answers**

🧩 Core Value Proposition
AI that only answers from your documents — fast, cited, and safe.

System Architecture

High-level Architecture



Architecture Table

Layer	Technology	Purpose
Frontend	React 18 + Typescript	UI & Streaming Chat
Backend	FastAPI	Async AI APIs
LLM	Groq	Ultra-fast interface
Vector DB	Pinecone	Semantic Retrieval
Embeddings	HuggingFace	Text embeddings
Cache	Redis	Sessions & speed

⚙️ Architecture Pattern
Pattern: Three-Tier Architecture
Deployment: Dockerized Microservices
Streaming: Server-Sent Events (SSE)

Feature Breakdown

Core Features

- ▼ Feature 1 - Document Management
  - ▼ Capability
    - PDF Upload (max 50MB)
    - Intelligent chunking
    - Metadata preservation
  - ▼ Technical Details
    - Chunk size: `1000 chars`
    - Overlap: `200 chars`
    - Vector dimension: `384`
  - ▼ Why it matters

    Enables precise retrieval and accurate citations.
- ▼ Feature 2 - AI Chat (Streaming)

  > ⚡ Sub-500ms Time To First Token (TTFT)

  - Real-time token streaming
  - Multi-turn context memory
  - Follow-up question reformulation
- ▼ Feature 3 - Citation System
  - Page number
  - Chunk index
  - Relevance score (0–100%)
  - Content preview
  - Source document name
- ▼ Feature 4 - Safety & Security

- ☑ ~~Hallucination detection~~
- ☑ ~~Scope enforcement~~
- ☑ ~~Rate limiting~~
- ☑ ~~Input sanitization~~

> 🚫 If information is not in documents → AI refuses to answer

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RAG Engine

1. Query Classification
2. History-Aware Reformulation
3. Hybrid Retrieval (70% semantic + 30% keyword)
4. Context Ranking
5. Retrieval Quality Analysis
6. LLM Response Generation (Groq)
    a. Safety Validation
7. Memory Storage
8. Streaming Delivery

> 💡 This multi-layer pipeline prevents hallucinations and ensures relevance at scale.

API Documentation

Method	Endpoint	Purpose
POST	/documents/upload	Upload PDFs

DELETE	/documents/{name}	Remove document
POST	/chat/stream	Streaming AI chat
GET	/health	System health

Deployment Guide

Docker Setup

```
docker-compose -f docker-compose.prod.yaml up --build
```

Environment Variables

```
# DocuMind Enterprise Configuration

# Groq Configuration (for AI chat)
GROQ_API_KEY=your_groq_api_key_here

# Pinecone Configuration (for vector storage)
PINECONE_API_KEY=your_pinecone_api_key_here
PINECONE_ENVIRONMENT=us-east-1
PINECONE_INDEX_NAME=documind-hf

# Embedding Configuration
EMBEDDING_PROVIDER=huggingface
EMBEDDING_MODEL=sentence-transformers/all-MiniLM-L6-v2

# Optional Settings
CHUNK_SIZE=1000
CHUNK_OVERLAP=200
MAX_FILE_SIZE_MB=50
```

Performance & Benchmarks

Performance Metrics

TTFT: `<500ms`

Tokens/sec: `30–50`

Search latency: `200–500ms`

50+ concurrent users

99% uptime (testing)

95% citation accuracy

Security & Compliance

Security & AI Safety

- Session isolation
- No external knowledge
- Audit logs
- API rate limiting

> ⚠️ Designed for enterprise compliance and audit readiness

Roadmap

▼ Short-Term (3 months)

- Multi-file formats
- Auth & roles
- Usage analytics

▼ Mid-Term (6 months)

- Multi-language
- SharePoint integration

▼ Long-Term (12+ months)

- Multi-modal RAG

- BI dashboards
- Enterprise SSO

Conclusion

DocuMind Enterprise demonstrates how modern AI systems must be built for real-world use — fast, grounded, secure, and auditable.

Unlike basic chatbot wrappers, this system enforces strict document-only reasoning, delivers real-time streaming responses, and provides verifiable citations for every answer.

With its modular architecture, production-grade safeguards, and clear roadmap for expansion, DocuMind Enterprise is not a proof-of-concept, but a scalable foundation for enterprise document intelligence.

💡 Built with production, not demos, in mind.